No.

Date

傅立叶变换　　数学意义

1. 几何→线代

三维空间　O-xyz.

① i坐标系　基向量 $\vec{i}$ $\vec{j}$ $\vec{k}$　两两正交.长度归一.

$\vec{OP} = x_p\vec{i} + y_p\vec{j} + z_p\vec{k}$　　$\vec{u_i}$

② j坐标系　基向量 $\vec{i'}$ $\vec{j'}$ $\vec{k'}$　$v_{ij,k}$　j系下k基向量在i系下坐标值.

$\vec{OP} = x_{p'}\vec{i'} + y_{p'}\vec{j'} + z_{p'}\vec{k'}$　　$\vec{u_j}$　　$u_{j,k}$ 坐标值.

$$\begin{cases} x_{p'} = \vec{OP}\cdot\vec{i'} \\ y_{p'} = \vec{OP}\cdot\vec{j'} \\ z_{p'} = \vec{OP}\cdot\vec{k'} \end{cases}$$

$\vec{u_i}^T = \sum_k u_{jk} v_{ijk}$　线性组合

$u_{j,k} = \sum_k \vec{u_i}^T \cdot v_{ijk}$.　投影

$\Rightarrow$　$u_j^T = u_i^T \cdot v_{ij}$　　原理:正交矩阵的转置是它的逆.

$u_i = v_{ij}\cdot u_j$　　$v_{ij}^{-1} = v_{ij}^T$.

2. 数组→连续函数的内积

① k维向量 $\vec{u}$　$\vec{v_i}$　求投影

$u(v_i) = u_1v_{i1} + u_2v_{i2} + \cdots + u_kv_{ik} = \sum_{n=1}^{k} u_nv_{in}$

② 数组 $f(n)=n^2+n$.　→不断减小 n 间隔.

→函数 $f(x)=x^2+x$　无穷维数组.

$\langle f(x), g(x)\rangle = \int_{-\infty}^{+\infty} f(x)\cdot g(x)\,dx$　函数内积-两个无穷维向量投影

3. 坐标变换→积分变换.

① $\underbrace{\langle f(x), g(x)\rangle}_{\text{值}} = \int_{-\infty}^{+\infty} \underbrace{f(x)}_{\text{原函数}}\cdot\underbrace{g(x)}_{\text{基函数}}\,dx$.

$\Rightarrow$　$\underbrace{F(w)}_{\text{像函数}} = \int_{-\infty}^{+\infty} \underbrace{f(x)}_{\text{原函数}}\cdot\underbrace{g(x,w)}_{\text{正交函数组}}\,dx$.

$\underbrace{f(x)}_{\text{原函数}} = \int_{-\infty}^{+\infty} \underbrace{F(w)}_{\text{像函数(坐标集合)}}\cdot g(x,w)\,dx$.

4. ② 在复数域内.　$\langle\vec{a},\vec{b}\rangle = \vec{a}\cdot\vec{b}^*$ (共轭).

$\Rightarrow$ $F(w) = \int_{-\infty}^{+\infty} f(x)\cdot g^*(x,w)\,dx$　　$f(x) = \int_{-\infty}^{+\infty} F(w)\cdot g(x,w)\,dx$

内积投影.　　线性组合

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扫描全能王 创建

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傅立叶变换.

1.三角函数的正交性.

① 三角函数系 $\{\underset{\sin 0x}{0}, \underset{\cos 0x}{1}, \sin x, \cos x, \cdots, \sin(nx), \cos(nx)\}$

② 正交性：任意两者乘积在 $[-\pi,\pi]$ 内积分为0.

$\int_{-\pi}^{\pi} \cos(nx)\,dx = \int_{-\pi}^{\pi} \cos(nx)\cdot 1\,dx = 0$ $\quad n=1,2,3,\cdots$

$\int_{-\pi}^{\pi} \sin(nx)\,dx = \int_{-\pi}^{\pi} \sin(nx)\cdot 1\,dx = 0$ $\quad n=1,2,3,\cdots$

$\int_{-\pi}^{\pi} \sin(kx)\cdot\cos(nx)\,dx = 0$ $\quad n,k=1,2,3,\cdots$

$\int_{-\pi}^{\pi} \cos(kx)\cdot\cos(nx)\,dx = 0$ $\quad n,k=1,2,3,\cdots\ n\neq k.$

$\int_{-\pi}^{\pi} \sin(kx)\cdot\sin(nx)\,dx = 0$ $\quad n,k=1,2,3,\cdots\ n\neq k$

证明：(1) $\int_{-\pi}^{\pi} \cos(nx)\,dx = \frac{1}{n}\sin(nx)\Big|_{-\pi}^{\pi} = \frac{1}{n}(0-0)=0.$

(2) $\int_{-\pi}^{\pi} \sin(nx)\,dx = -\frac{1}{n}\cos(nx)\Big|_{-\pi}^{\pi}$ (偶函数) $=0.$

积化和差

(3) $\int_{-\pi}^{\pi} \sin(kx)\cos(nx) = \frac{1}{2}\int_{-\pi}^{\pi}[\sin(kx+nx)+\sin(kx-nx)]\,dx = 0.$

(4) $\int_{-\pi}^{\pi} \cos(kx)\cos(nx) = \frac{1}{2}\int_{-\pi}^{\pi}[\cos(kx+nx)+\cos(kx-nx)]\,dx = 0.$

(5) $\int_{-\pi}^{\pi} \sin(kx)\cdot\sin(nx) = -\frac{1}{2}\int_{-\pi}^{\pi}[\cos(kx+nx)-\cos(kx-nx)]\,dx = 0.$

$\int_{-\pi}^{\pi} \sin(nx)\sin(nx)\,dx = \pi$ $\quad n=1,2,3,\cdots$

$\int_{-\pi}^{\pi} \cos(nx)\cos(nx)\,dx = \pi$ $\quad n=1,2,3,\cdots$

证明：(1) $\int_{-\pi}^{\pi} \sin^2(nx)\,dx = \frac{1}{2}\int_{-\pi}^{\pi}[1-\cos(2nx)]\,dx = \pi.$

$\int_{-\pi}^{\pi} \cos^2(nx)\,dx = \frac{1}{2}\int_{-\pi}^{\pi}[1+\cos(2nx)]\,dx = \pi.$

2.周期为2π的函数展开为傅立叶级数.

$f(x) = f(x+2\pi) \qquad T=2\pi.$

$\Rightarrow f(x) = \sum_{n=0}^{\infty} a_n\cos(nx) + \sum_{0}^{\infty} b_n\sin(nx)$ ①

$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}[a_n\cos(nx) + b_n\sin(nx)]$ ②.

推导：① $\int_{-\pi}^{\pi} f(x)\,dx = \underbrace{\int_{-\pi}^{\pi} a_0}_{2\pi a_0} + \underbrace{\int_{-\pi}^{\pi}\sum_{n=1}^{\infty} a_n\cos(nx)\,dx}_{0} + \underbrace{\int_{-\pi}^{\pi}\sum_{n=1}^{\infty} b_n\sin(nx)\,dx}_{0}.$

$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$

② 令 $a_0(new) = \frac{a_0}{2}$ 则 $a_0(new) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$ (形式统一).

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积化和差

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$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}[a_n\cos(nx) + b_n\sin(nx)]$ ②.

推导: ①. $\int_{-\pi}^{\pi} f(x)\,dx = \underbrace{\int_{-\pi}^{\pi} a_0}_{2\pi a_0} + \underbrace{\int_{-\pi}^{\pi}\sum_{n=1}^{\infty} a_n\cos(nx)\,dx}_{0} + \underbrace{\int_{-\pi}^{\pi}\sum_{n=1}^{\infty} b_n\sin(nx)\,dx}_{0}.$

$a_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\,dx$

②. 令 $a_0(new) = \frac{a_0}{2}$ 则 $a_0(new) = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$ (形式统一).

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傅立叶变换.

$$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\cos(nx)\,dx.$$

$$b_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\sin(nx)\,dx.$$

推导: $\int_{-\pi}^{\pi}f(x)\cos(mx)\,dx$.

$$=\underbrace{\int_{-\pi}^{\pi}\frac{a_0}{2}\cos(mx)\,dx}_{0}+\underbrace{\int_{-\pi}^{\pi}\sum_{n=1}^{\infty}a_n\cos(nx)\cos(mx)\,dx}_{n\neq m\ 0\quad n=m\ \pi a_n}+\underbrace{\int_{-\pi}^{\pi}\sum_{n=1}^{\infty}b_n\sin(nx)\cos(mx)\,dx}_{0}$$

$$\Rightarrow a_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\cos(nx)\,dx.$$

3. 周期为2L的函数展开为傅立叶级数

~~$f(x)=f(x+2L)$~~ $T=2L$. $f(t)=f(t+2\pi)$.

换元: $x=\frac{\pi}{L}t$.

$t=\frac{L}{\pi}x$.

$f(t)=f(\frac{L}{\pi}x)\Rightarrow g(x)$.

$$\Rightarrow\begin{cases}\cos nx=\cos\frac{n\pi}{L}t.\\ \sin nx=\cos\frac{n\pi}{L}t\\ \frac{1}{\pi}\int_{-\pi}^{\pi}dx=\frac{1}{L}\int_{-L}^{L}dt.\end{cases}$$

$$\Rightarrow\begin{cases}g(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}[a_n\cos(nx)+b_n\sin(nx)].\\ a_0=\frac{1}{\pi}\int_{-\pi}^{\pi}g(x)\,dx.\\ a_n=\frac{1}{\pi}\int_{-\pi}^{\pi}g(x)\cos(nx)\,dx.\\ b_n=\frac{1}{\pi}\int_{-\pi}^{\pi}g(x)\sin(nx)\,dx.\end{cases}\Rightarrow\begin{cases}f(t)=\frac{a_0}{2}+\sum_{n=1}^{\infty}[a_n\cos\frac{n\pi}{L}t+b_n\sin\frac{n\pi}{L}t]\\ a_0=\frac{1}{L}\int_{-L}^{L}f(t)\,dt\\ a_n=\frac{1}{L}\int_{-\pi}^{\pi}f(t)\cos(\frac{n\pi}{L}t)\,dt\\ b_n=\frac{1}{L}\int_{-\pi}^{\pi}f(t)\sin(\frac{n\pi}{L}t)\,dt.\end{cases}$$

工程中: t从0开始 $T=2L$. $w=\frac{\pi}{L}=\frac{2\pi}{T}$.

$$\int_{-L}^{L}dt\to\int_{0}^{2L}dt\to\int_{0}^{T}dt.$$

$$f(t)=\frac{a_0}{2}+\sum_{n=1}^{\infty}[a_n\cos(nwt)+b_n\sin(nwt)].$$

$$a_0=\frac{2}{T}\int_{0}^{T}f(t)\,dt.$$

$$a_n=\frac{2}{T}\int_{0}^{T}f(t)\cos(nwt)\,dt.$$

$$b_n=\frac{2}{T}\int_{0}^{T}f(t)\sin(nwt)\,dt.$$

4. 傅立叶级数的复数形式.

欧拉公式 $e^{i\theta}=\cos\theta+i\sin\theta$.

$$\Rightarrow\begin{cases}\cos\theta=\frac{1}{2}(e^{i\theta}+e^{-i\theta})\\ \sin\theta=\frac{1}{2i}(e^{i\theta}-e^{-i\theta}\end{cases}$$

复数形式 $f(t)=f(t+T)$.

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傅立叶变换 3.

$$f(t)=\sum_{-\infty}^{\infty} C_n e^{in\omega t}.$$

$$C_n=\frac{1}{T}\int_0^T f(t)e^{-in\omega t}dt.$$

推导: $f(t)=\frac{a_0}{2}+\sum_{n=1}^{\infty}\left[a_n\frac{1}{2}(e^{in\omega t}+e^{-in\omega t})-\frac{1}{2}ib_n(e^{in\omega t}-e^{-in\omega t})\right].$

$$=\frac{a_0}{2}+\sum_{n=1}^{\infty}\left[\frac{a_n-ib_n}{2}e^{in\omega t}+\frac{a_n+ib_n}{2}e^{-in\omega t}\right].$$

$$=\frac{a_0}{2}+\sum_{n=1}^{\infty}\frac{a_n-ib_n}{2}e^{in\omega t}+\sum_{n=1}^{\infty}\frac{a_n+ib_n}{2}e^{-in\omega t}. \qquad n=\text{"}-n\text{"}$$

$$=\sum_{n=0}^{0}\frac{a_0}{2}e^{in\omega t}+\sum_{n=1}^{\infty}\frac{a_n-ib_n}{2}e^{in\omega t}+\sum_{n=-\infty}^{-1}\frac{a_{-n}+ib_{-n}}{2}e^{in\omega t}.$$

$$=\sum_{-\infty}^{\infty}C_n e^{in\omega t}$$

$$\Rightarrow \quad C_n=\begin{cases}\frac{a_0}{2} & n=0\\ \frac{a_n-ib_n}{2} & n=1,2,3,\cdots\\ \frac{a_{-n}+ib_{-n}}{2} & n=-1,-2,-3,\cdots\end{cases}$$

$n=0$ $\quad C_n=\frac{a_0}{2}=\frac{1}{2}\cdot\frac{2}{T}\int_0^T f(t)dt=\frac{1}{T}\int_0^T f(t)dt=\frac{1}{T}\int_0^T f(t)e^{-in\omega t}dt.$

$n=1,2,3\cdots$ $\quad C_n=\frac{1}{2}\left(\frac{2}{T}\int_0^T f(t)\cos(n\omega t)dt-i\frac{2}{T}\int_0^T f(t)\sin(n\omega t)dt\right).$

$$=\frac{1}{T}\int_0^T f(t)\left[\underbrace{\cos(n\omega t)-i\sin(n\omega t)}_{\cos(-n\omega t)+i\sin(-n\omega t)\to e^{-in\omega t}}\right]dt.$$

$$=\frac{1}{T}\int_0^T f(t)e^{-in\omega t}dt.$$

$n=-1,-2,\cdots$ $\quad C_n=\frac{1}{2}\left(\frac{2}{T}\int_0^T f(t)\cos(-n)\omega t\,dt+i\frac{2}{T}\int_0^T f(t)\sin(-n)\omega t\,dt\right).$

$$=\frac{1}{T}\int_0^T f(t)\left[\cos(-n\omega t)+i\sin(-n\omega t)\right]dt.$$

$$=\frac{1}{T}\int_0^T f(t)e^{-in\omega t}dt.$$

$$\Rightarrow \quad C_n=\frac{1}{T}\int_0^T f(t)e^{-in\omega t}dt.$$

5. 傅立叶变换.

$f(t)=f(t+T)$ $\qquad \omega_0=\frac{2\pi}{T}$ 基频率.

$f(t)=\sum_{-\infty}^{+\infty}C_n e^{in\omega_0 t}.$ $\qquad$ (1) 时域

$C_n=\frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}}f(t)e^{-in\omega_0 t}dt$ $\qquad$ (2). 频域 → 频谱

$\Rightarrow$ $T\to\infty.$ $\quad \Delta\omega=\frac{2\pi}{T}\to d\omega.$ $\qquad \frac{1}{T}=\frac{\Delta\omega}{2\pi}\to\frac{d\omega}{2\pi}.$

(2)代入(1)得 $\quad f_T(t)=\underbrace{\sum_{n=-\infty}^{+\infty}\frac{1}{T}}_{\int_{-\infty}^{+\infty}\frac{d\omega}{2\pi}}\underbrace{\int_{-\frac{T}{2}}^{\frac{T}{2}}}_{\int_{-\infty}^{+\infty}}f(t)e^{-in\omega_0 t}dt\,e^{in\omega_0 t}.$

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傅立叶变换.

$$f(t)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty}f(t)e^{-i\omega t}dt\,e^{i\omega t}d\omega$$

$$\begin{cases}\text{傅立叶变换} & F(\omega)=\int_{-\infty}^{+\infty}f(t)e^{-i\omega t}dt. \quad \text{FT}\\ \text{傅立叶逆变换} & f(t)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}F(\omega)e^{i\omega t}d\omega. \quad \text{IFT}\end{cases}$$

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